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Effects of impurities on zonal flows in tokamak plasmas

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ABSTRACT

Effects of impurities on zonal flows (ZFs) driven by the ion density and temperature perturbations in tokamak plasmas are investigated using gyrokinetic theory and simulations. The Rosenbluth–Hinton collisionless gyrokinetic model for ZFs is used, with model source terms designed to represent the drive from different plasma species in the plasma. For ZFs driven by the main ion density perturbations, the presence of impurities could decrease the amplitude of ZFs by increasing the total ion mass density of plasmas. For ZFs driven by the main ion temperature perturbations, the presence of impurities could decrease the amplitude of ZFs by decreasing the main ion mass density fraction. In addition, the driving effects of the impurity ion temperature perturbations on ZFs can be ignored, due to the low concentration of impurities.

I. INTRODUCTION

Zonal flows (ZFs) play an important role in tokamak plasmas since they are believed to regulate drift wave turbulence,[1,2] reduce anomalous transport,[3] and improve plasma confinement, e.g., the transition from low- to high-confinement (L–H transition).[4] Different from the background $\boldsymbol{E} \times \boldsymbol{B}$ flow, which is driven by the sheared radial electric field in the edge region and responsible for L–H transition, ZFs exhibit a smaller-scale oscillatory structure in the minor-radial direction. It is found that linearly stable mesoscale ZFs with long-wavelength ($k_r \rho_{\mathrm{L,i}} \ll 1$) radial structure can be generated spontaneously in ion temperature-gradient (ITG) turbulence.[3] Here, k_r denotes the radial wavenumber of ZF and $\rho_{\mathrm{L,i}}$ denotes the ion Larmor gyroradius. Rosenbluth and Hinton[5] (RH) demonstrated that ZFs driven by ITG turbulence are not damped by linear collisionless processes but are modified by the neoclassical polarization shielding factor, $\epsilon_r = 1/\left(1 + 1.6q^2/\sqrt{\epsilon}\right)$, in toroidal geometry. Here, q is the safety factor and $\epsilon = r/R$ is the inverse aspect-ratio of the tokamak, with r and R being the minor and major radii, respectively. Subsequently, numerous theories[6–10] and simulations[11–13] for ZFs nonlinearly driven by drift wave turbulence in toroidal geometry are carried out. From a microscopic viewpoint, ZFs can be excited through modulational instability[6] and eigenmode self-interaction.[11,12] From a mesoscopic viewpoint, ZFs can be driven by turbulent poloidal Reynolds stress,[7–9] turbulent toroidal Reynolds stress, and turbulent energy flux.[10,13]

All the works mentioned above investigate ZFs in pure plasma. In fact, impurities are ubiquitous in tokamak plasmas due to the existence of the plasma–wall interactions and the fusion reactions.[14] To investigate the effects of impurities on ZFs, several theories[15,16] and simulations[17–19] have been performed. Linear gyrokinetic theories[15,16] show that the presence of impurities has very weak influence on the ratio of the residual to the initial zonal perturbed electrostatic potential, $\delta\phi(\infty)/\delta\phi(0)$, driven by initial charge density perturbations in the long-wavelength limit and the collisionless case, and this theoretical prediction has been verified by linear collisionless gyrokinetic simulations.[17,18] In addition, using the generalized multi-species Hasegawa–Wakatani model, strongly enhanced perturbations of heavy impurity density are found in long-living plasma flow vortices.[19] Moreover, the nonlinear collisionless gyrokinetic simulation[18] shows that the presence of impurities could reduce the time-averaged absolute value of ZFs in the ITG turbulence.

Note that the present linear gyrokinetic theories[15,16] and simulations[17–19] generally investigate the effects of impurities on ZFs driven by charge density perturbations. Experimental observations of limit-cycle-oscillations[20,21] indicate that the pressure gradient plays an essential role in L–H transition. Further gyrokinetic theory[9,10] and simulations[12,13] indicate that the temperature perturbations induced by turbulent energy flux play an important role in driving ZFs. In contrast to charge density perturbations, the driving effects of temperature perturbations are not shielded by neoclassical polarization in toroidal geometry.[9,10] Therefore, it is important to investigate the effects of impurities on ZFs driven not only by charge density perturbations but also by temperature perturbations.

In this work, we investigate the effects of impurities on ZFs driven by ion density and temperature perturbations in tokamak plasmas. It is found that ion density and temperature perturbations represent the contributions from turbulent poloidal Reynolds stress[9] and turbulent energy flux,[10] respectively. The Rosenbluth–Hinton collisionless gyrokinetic model for ZFs is used, with model source terms designed to represent the drive from different plasma species in the plasma. The results show that the presence of impurities significantly reduces the amplitude of mesoscale ZFs driven by the main ion density and temperature perturbations. For ZFs driven by the main ion density perturbations, the presence of impurities has little effect on the ratio of the residual to the initial zonal potential, $\delta\phi(\infty)/\delta\phi(0)$; however, it could decrease the amplitude of ZFs by increasing the total ion mass density of plasmas. On the other hand, for ZFs driven by the main ion temperature perturbations, the presence of impurities could decrease the amplitude of ZFs by decreasing the main ion mass density fraction. Finally, the driving effects of impurity ion temperature perturbations on ZFs can be ignored, due to the low concentration of impurities. Note that collisionless assumption is adopted in this paper, similar to Refs. 16–18, where the effects of impurities on ZFs are discussed. However, it should be pointed out that collisional damping can reduce the amplitude of ZFs in a pure plasma[22] or in a plasma with impurity.[15] Therefore, the conclusion in this paper should be understood as the upper limit for the amplitude of ZFs driven by turbulent poloidal Reynolds stress and turbulent energy flux in ITG turbulence in plasmas with impurities.

The remaining parts of this paper are organized as follows: In Sec. II, we present the gyrokinetic theory of ZFs in the presence of impurities and discuss the mechanism of impurity-induced effects on ZFs; in Sec. III, we present the gyrokinetic simulation results and compare them with theoretical predictions; and in Sec. IV, we summarize the main conclusions.

II. GYROKINETIC THEORY OF ZONAL FLOWS DRIVEN BY DENSITY AND TEMPERATURE PERTURBATIONS IN TOROIDAL PLASMAS WITH IMPURITIES

To investigate the mesoscale dynamics of ZFs, following Ref. 5, we begin with the ensemble-averaged Frieman–Chen gyrokinetic equation[23] for the zonal part of the perturbed distribution function,

$$\partial_t g_j + \mathcal{L}_{0,j} g_j = \frac{e_j}{T_j} F_{0,j} J_0 \partial_t \delta\phi + J_0 S_j F_{0,j}, \tag{1}$$

with $j = \mathrm{i}, \mathrm{z}, \mathrm{e}$ denoting main ions, impurity ions, and electrons, respectively. $\mathcal{L}_{0,j} \equiv (v_\parallel \boldsymbol{b} + \boldsymbol{V}_{\mathrm{d},j}) \cdot \nabla$, with $v_\parallel$ being the parallel velocity and $\boldsymbol{b} = \boldsymbol{B}/B$. $\boldsymbol{B}$ is the equilibrium magnetic field. $\boldsymbol{V}_{\mathrm{d},j}$ is the magnetic drift velocity in the equilibrium magnetic field. The perturbed distribution function for ZF is $\delta f_j = -(e_j \delta\phi/T_j) F_{0,j} + g_j(\boldsymbol{X}, K, \mu, t)$. The equilibrium distribution function is taken as the local Maxwellian $F_{0,j} = \pi^{(-3/2)} v_{\mathrm{th},j}^{-3} n_j \exp(-K/T_j)$, with $m_j v_{\mathrm{th},j}^2/2 = T_j$. n_j and T_j are the equilibrium density and temperature, respectively. m_j is the particle mass. $K = \frac{1}{2} m_j v_\parallel^2 + \mu B$, with μ being the magnetic moment. $\delta\phi$ is the zonal perturbed electrostatic potential, which is assumed to be given by the eikonal form $\delta\phi(\boldsymbol{x}) = \delta\phi_k \exp(\mathrm{i} k_r r)$. $\boldsymbol{X} = \boldsymbol{x} - \boldsymbol{\rho}_{\mathrm{L},j}$ is the guiding center position, with $\boldsymbol{x}$ being the particle position and $\boldsymbol{\rho}_{\mathrm{L},j} = \boldsymbol{b} \times m_j \boldsymbol{v}/e_j B$. $\boldsymbol{v}$ and e_j are the particle velocity and charge, respectively. The zeroth-order Bessel function $J_0 = J_0(k_r \rho_{\mathrm{L},j})$, which represents the gyroaverage.

The source term S_j in Eq. (1), which represents the driving effects of ITG turbulence on ZFs, is given by[10]

$$S_j F_{0,j} = -\frac{1}{r} \partial_r \left(r \langle \tilde{V}_r \tilde{f}_j \rangle_{\mathrm{en}} \right), \tag{2}$$

with $\tilde{V}_r$ and $\tilde{f}_j$ representing the radial component of the turbulent fluctuating $\boldsymbol{E} \times \boldsymbol{B}$ velocity and the turbulent fluctuating distribution function, respectively. $\langle\rangle_{\mathrm{en}}$ represents ensemble average. Following Refs. 9 and 10, one can decompose the phase-space turbulent transport flux by introducing the moment approach, which yields

$$S_j = -\frac{1}{r} \partial_r (r \Gamma_{r,j}^{\mathrm{tot}}) \frac{1}{n_j} - \frac{2}{3} \frac{1}{r} \partial_r \left[r \left(\frac{K}{T_j} - \frac{3}{2} \right) q_{r,j} \right] \frac{1}{p_j}, \tag{3}$$

with $\Gamma_{r,j}^{\mathrm{tot}} = \Gamma_{r,j} - \frac{1}{e_j B} \frac{1}{r} \partial_r (r \Pi_{r\theta,j})$ and $q_{r,j} = Q_{r,j} - \frac{3}{2} \Gamma_{r,j} T_j$ being the total turbulent particle flux and the turbulent heat flux, respectively. The turbulent toroidal Reynolds stress[10] is ignored here. Note that $\Gamma_{r,j}$ is the auto-ambipolar component of the total turbulent particle flux, while the non-ambipolar turbulent particle flux is given by turbulent poloidal Reynolds stress $\Pi_{r\theta,j}$.[9] The turbulent poloidal Reynolds stress $\Pi_{r\theta,j}$ and the turbulent energy flux $Q_{r,j}$ are expressed as

$$\Pi_{r\theta,j} = \left\langle \int \mathrm{d}^3 \boldsymbol{v} F_{0,j} m_j \tilde{V}_r \tilde{V}_\theta \right\rangle_{\mathrm{en}}, \tag{4}$$

$$Q_{r,j} = \left\langle \int \mathrm{d}^3 \boldsymbol{v} \tilde{V}_r \tilde{f}_j K \right\rangle_{\mathrm{en}}, \tag{5}$$

respectively, with $\tilde{V}_\theta$ being the poloidal component of the fluctuating $\boldsymbol{E} \times \boldsymbol{B}$ velocity.

Introducing[10]

$$-\frac{1}{r} \partial_r (r \Gamma_{r,j}^{tot}) = \partial_t \delta n_j, \tag{6}$$

$$-\frac{1}{r} \partial_r (r q_{r,j}) = \frac{3}{2} n_j \partial_t \delta T_j, \tag{7}$$

which can be found by taking the velocity moments of Eq. (1), with the $\mathcal{L}_{0,j}$ term and the zonal potential term ignored. δn_j and δT_j denote the density and temperature perturbations, which represent the turbulent poloidal Reynolds stress and the turbulent energy flux, respectively. Note that, since $m_\mathrm{i} \gg m_\mathrm{e}$, the driving effects of the ion turbulent poloidal Reynolds stress $\Pi_{r\theta,\mathrm{i}}$ on ZFs are much stronger than that of the electron turbulent poloidal Reynolds stress $\Pi_{r\theta,\mathrm{e}}$, which indicates that density perturbations have the non-auto-ambipolar effects.[9]

Combining Eqs. (3), (6), and (7), we have the source term, representing the driving effects of plasma turbulence on ZFs, written in the eikonal form $S_j = S_{j,k} \exp(\mathrm{i} k_r r)$ as follows:

$$S_{j,k} = \frac{\partial_t \delta n_{j,k}}{n_j} + \left(\frac{K}{T_j} - \frac{3}{2} \right) \frac{\partial_t \delta T_{j,k}}{T_j}. \tag{8}$$

$\delta n_{j,k}$ and $\delta T_{j,k}$ satisfy

$$\left[\delta n_j, \delta T_j \right] = \left[\delta n_{j,k}, \delta T_{j,k} \right] \mathrm{e}^{\mathrm{i} k_r r}. \tag{9}$$

Following Ref. 10, the initial impulse source is adopted for simplicity, which is written as

$$S_{j,k} = \delta(t)\left[\frac{\delta n_{j,k}(0)}{n_j} + \left(\frac{K}{T_j} - \frac{3}{2}\right)\frac{\delta T_{j,k}(0)}{T_j}\right], \quad (10)$$

with the delta function $\delta(t)$ understood as a function whose width is longer than a gyro-period but shorter than a bounce period.[5] We point out that the ZF driven by the nonlinear source given as a delta function in the RH test[5] should be understood as Green's function[10] of the problem; the delta function should not be simply understood as the initial condition.[10]

Following Refs. 5 and 10, Eq. (1) is solved by introducing $\partial_t \ll \mathcal{L}_{0,j} \sim 1/\tau_{\theta,j}$ for ZFs, with $\tau_{\theta,j}$ being the poloidal transit time for passing particles or the bounce time for trapped particles. Therefore, one can make the expansion $g_j = g_j^{(0)} + g_j^{(1)}$, with

$$\mathcal{L}_{0,j} g_j^{(0)} = 0, \quad (11)$$

$$\partial_t g_j^{(0)} + \mathcal{L}_{0,j} g_j^{(1)} = \frac{e_j}{T_j} F_{0,j} J_0 \partial_t \delta\phi + J_0 S_j F_{0,j}. \quad (12)$$

Equation (11) indicates that $g_j^{(0)}$ should be a constant of unperturbed motion. $g_j^{(0)}$ is determined by the solubility condition, the orbital average of Eq. (12).[10] Assuming $g_j^{(0)} = g_{j,k}\exp(\mathrm{i}k_r r)$, $g_{j,k}$ can be written as

$$g_{j,k} = e^{-\mathrm{i}Q_j}\left[\frac{e_j}{T_j}\overline{\left(e^{\mathrm{i}Q_j} J_0 \delta\phi_k\right)} + \overline{\left(e^{\mathrm{i}Q_j} J_0 R_{j,k}\right)}\right] F_{0,j}. \quad (13)$$

Here, $Q_j = k_r(r - \overline{r})$, with $\overline{r} = \frac{1}{\tau_{\theta,j}}\oint \mathrm{d}\theta \frac{1}{(v_\parallel \boldsymbol{b} + \boldsymbol{V}_{\mathrm{d},j})\cdot\nabla\theta} r$. $R_{j,k} = \int \mathrm{d}t S_{j,k}$. Note that the distribution function for electrons is given by setting $J_0(k_r\rho_{\mathrm{L,e}}) = 1$ and $Q_{\mathrm{e}} = 0$ due to the much smaller Larmor radius and banana width of electrons.

The zonal potential $\delta\phi_k$ is determined by the magnetic flux-surface averaged quasi-neutrality equation including main ions, impurity ions, and electrons, which is written as follows:

$$\sum_{j=\mathrm{i,z,e}} e_j \left\langle -\frac{e_j}{T_j} n_j \delta\phi_k + \int \mathrm{d}^3\boldsymbol{v} J_0(k_r\rho_{\mathrm{L},j}) g_{j,k}\right\rangle_{\mathrm{FA}} = 0. \quad (14)$$

Here, $\langle\rangle_{\mathrm{FA}}$ represents the magnetic flux-surface average. By adopting the long-wavelength approximation and ignoring the contribution of electrons to the source, the temporal asymptotic zonal potential $\delta\phi_k(\infty)$ can be solved from Eqs. (13) and (14), which yields

$$\epsilon_r \frac{\rho_{\mathrm{M}}}{B^2} k_r^2 \delta\phi_k(\infty) = \sum_{j=\mathrm{i,z}} e_j \delta n_{j,k}(0) - \epsilon_r \sum_{j=\mathrm{i,z}} \frac{\rho_{\mathrm{M},j}}{B^2} k_r^2 \frac{\delta T_{j,k}(0)}{e_j}. \quad (15)$$

Here, $\rho_{\mathrm{M}} = \sum_{j=\mathrm{i,z}} n_j m_j$ is the total ion mass density and $\rho_{\mathrm{M},j} = n_j m_j$ is the mass density of the ion species j. Note that, when the driving source is density perturbations, the asymptotic zonal potential $\delta\phi_k(\infty)$ should be understood as the residual zonal potential. The details for computing the magnetic flux-surface average of the velocity integral in Eq. (14) can be found in Refs. 9 and 10. From Eq. (15), one can find that the driving effects of temperature perturbations are enhanced by the same neoclassical polarization factor ϵ_r (the toroidal geometry effects) as the asymptotic zonal potential, while the driving effects of density perturbations are not. This indicates that the driving effects of temperature perturbations on ZFs are not shielded by the neoclassical polarization process in tokamak geometry, while the driving effects of density perturbations on ZFs are shielded, which is consistent with the results reported in previous works.[9,10]

Note that two quantities in Eq. (15) represent the effects of impurities on ZFs. The first is the total ion mass density, ρ_{M}, which is related to the classical dielectric constant of plasma. $\rho_{\mathrm{M}} = n_{\mathrm{i}} m_{\mathrm{i}}$ in pure plasma, while $\rho_{\mathrm{M}} = n_{\mathrm{i}} m_{\mathrm{i}} + n_{\mathrm{z}} m_{\mathrm{z}}$ in admixed plasma. The effect of impurities on ρ_{M} indicates that impurities are involved in the classical polarization process, and thus the presence of impurities can affect the amplitude of the zonal potential. The second is the impurity density perturbations δn_{z} and temperature perturbations δT_{z}, which represent the contributions from the impurity turbulent transport flux and also have driving effects on ZFs. Note that the left-hand side of Eq. (15) contains the linear polarization density of impurity, while the right-hand side of Eq. (15) contains the driving effect of impurity, which is given by the impurity's gyrocenter distribution function, g_{z}. In a typical plasma with trace amount of impurities, the driving effects of impurities are weak, which will be discussed in the end of this section. Therefore, ignoring impurity collisional effects that affect g_{z}, similar to Refs. 16–18, is unlikely to change the main conclusion of this paper, although collisional effects may further reduce ZFs.[15,22] Note that impurities affect ZFs through modifying classical and neoclassical polarization, determined by impurity mass and represented in the left-hand side of Eq. (15). Therefore, the main conclusion of this paper does not change when partially ionized tungsten impurities are considered.

Equation (15) is the main result of this paper, which gives the ZF (asymptotic value) driven by density and temperature perturbations. In order to compare with the previous theory,[16] which gives the impurity effects on the ratio of the asymptotic value to the initial value of ZF, we need to consider the initial zonal potential excited by initial perturbations. Note that initial ion temperature perturbations will not generate the initial zonal potential, while initial ion density perturbations will generate the initial zonal potential $\delta\phi_k(0)$ by the classical polarization process, which yields

$$\frac{c^2}{V_{\mathrm{A}}^2}\varepsilon_0 k_r^2 \delta\phi_k(0) = \sum_{j=\mathrm{i,z}} e_j \delta n_{j,k}(0). \quad (16)$$

Here, $V_{\mathrm{A}} = B/\sqrt{\mu_0\rho_{\mathrm{M}}}$ and c are the Alfvén speed and the speed of light in free space, respectively. μ_0 and ε_0 are the magnetic permeability and the dielectric constant in free space, respectively. c^2/V_{A}^2 represents the classical relative dielectric constant of plasma, which describes the classical polarization process. By combining Eqs. (15) and (16), one finds that

$$\delta\phi_k(\infty) = \frac{\delta\phi_k(0)}{\epsilon_r} - \sum_{j=\mathrm{i,z}} \frac{\rho_{\mathrm{M},j}}{\rho_{\mathrm{M}}} \frac{\delta T_{j,k}(0)}{e_j}. \quad (17)$$

Here, $\rho_{\mathrm{M},j}/\rho_{\mathrm{M}}$ denotes the mass density fraction of ion species j. In pure plasma, $\rho_{\mathrm{M,i}}/\rho_{\mathrm{M}} = 1$, and thus the present theory [Eq. (17)] recovers the previous theory[10] on the driving effects of main ions.

In the presence of impurities, Eq. (17) shows that ZFs driven by initial main ion density perturbations, $\delta n_{\mathrm{i}}(0)$, are shielded by the same neoclassical polarization shielding factor, ϵ_r, as that in pure plasma. This indicates that the presence of impurities has no effect on the neoclassical polarization shielding factor, which is consistent with the previous gyrokinetic theory[16] in the long-wavelength limit. Here, we point out that Eq. (16) shows that the value of the initial zonal potential

$\delta\phi(0)$ is related to the total ion mass density, ρ_{M}, which is included in the classical relative dielectric constant, c^2/V_{A}^2, and can be affected by impurities. This indicates that the presence of impurities affect $\delta\phi(0)$. In order to further estimate this effect quantitatively, we use the quasi-neutrality condition $\sum_{j=\mathrm{i,z,e}} e_j n_j = 0$ and adopt $e_{\mathrm{i}} = -e_{\mathrm{e}}$ to rewrite the total ion mass density,

$$\rho_{\mathrm{M}} = n_i m_i + n_z m_z \tag{18}$$

$$= n_{\mathrm{e}}\left[m_{\mathrm{i}} + e_{\mathrm{z}}\frac{n_{\mathrm{z}}}{n_{\mathrm{e}}}\left(\frac{1}{e_{\mathrm{z}}/m_{\mathrm{z}}} - \frac{1}{e_{\mathrm{i}}/m_{\mathrm{i}}}\right)\right]. \tag{19}$$

Equation (19) shows that, if the electron density n_{e} is fixed and the charge-to-mass ratio of the impurity ion, $e_{\mathrm{z}}/m_{\mathrm{z}}$, is smaller than that of the main ion, $e_{\mathrm{i}}/m_{\mathrm{i}}$, the presence of impurities increases the value of the total ion mass density, ρ_{M}. This indicates the following: (1) The presence of impurities will increase the classical relative dielectric constant, c^2/V_{A}^2, and will eventually decrease the amplitude of the asymptotic zonal potential $\delta\phi(\infty)$ driven by the main ion density perturbations. (2) The higher the impurity concentration $n_{\mathrm{z}}/n_{\mathrm{e}}$, the lower the amplitude of the asymptotic zonal potential $\delta\phi(\infty)$. Note that this mechanism is similar to that of impurity-induced effects on geodesic acoustic modes (GAMs), in which impurities affect the classical polarization process by the effective ion mass number, and thus affect the GAM behavior.[24,25]

For ZFs driven by initial main ion temperature perturbations, $\delta T_{\mathrm{i}}(0)$, Eq. (17) shows that the amplitude of the asymptotic zonal potential $\delta\phi(\infty)$ is related to the main ion mass density fraction, $\rho_{\mathrm{M,i}}/\rho_{\mathrm{M}}$. Note that $\rho_{\mathrm{M,i}}/\rho_{\mathrm{M}} = 1$ in pure plasma, while $\rho_{\mathrm{M,i}}/\rho_{\mathrm{M}} < 1$ in admixed plasma, which indicates that the presence of impurities decreases the amplitude of the asymptotic zonal potential $\delta\phi(\infty)$. Furthermore, if one writes $\rho_{\mathrm{M,i}}/\rho_{\mathrm{M}} = \frac{m_{\mathrm{i}}}{m_{\mathrm{i}} + m_z n_z/n_{\mathrm{i}}}$, it can be found that when the impurity ion density n_{z} increases, the main ion mass density fraction $\rho_{\mathrm{M,i}}/\rho_{\mathrm{M}}$ decreases, and thus the decreasing effects of impurities on the amplitude of the asymptotic zonal potential $\delta\phi(\infty)$ are enhanced.

Finally, we point out that the driving effects of impurity ion temperature perturbations, $\delta T_{\mathrm{z}}(0)$, on ZFs can be ignored. This can be demonstrated as follows. It is found that $T_{\mathrm{z}} \sim T_{\mathrm{i}}$ in tokamaks, such as the DIII-D.[26] Assuming $\delta T_j(0) \propto T_j$ in turbulence, one finds $\delta T_{\mathrm{z}}(0) \sim \delta T_{\mathrm{i}}(0)$. If one rewrites Eq. (17) as

$$\delta\phi_k(\infty) = \frac{\delta\phi_k(0)}{\epsilon_r} - \frac{1}{\rho_{\mathrm{M}}}\sum_{j=\mathrm{i,z}} n_j \frac{m_j}{e_j}\delta T_{j,k}(0), \tag{20}$$

it can be seen that since $m_{\mathrm{z}}/e_{\mathrm{z}} \sim m_{\mathrm{i}}/e_{\mathrm{i}}$ while $n_{\mathrm{z}} \ll n_{\mathrm{i}}$ in typical tokamak plasmas, the driving effects of impurity ion temperature perturbations $\delta T_{\mathrm{z}}(0)$ on ZFs are much weaker than that of the main ion temperature perturbations $\delta T_{\mathrm{i}}(0)$. Therefore, the driving effects of impurity ion temperature perturbations can be ignored, which yields

$$\delta\phi_k(\infty) = \frac{\delta\phi_k(0)}{\epsilon_r} - \frac{1}{\rho_{\mathrm{M}}} n_{\mathrm{i}} \frac{m_{\mathrm{i}}}{e_{\mathrm{i}}}\delta T_{\mathrm{i},k}(0). \tag{21}$$

III. GYROKINETIC SIMULATIONS OF ZONAL FLOWS DRIVEN BY THE DENSITY AND TEMPERATURE PERTURBATIONS IN TOROIDAL PLASMAS WITH IMPURITIES

In Sec. II, the effects of impurities on ZFs driven by ion density and temperature perturbations are investigated using gyrokinetic theory. The results show that for ZFs driven by main ion density perturbations, the presence of impurities can increase the total ion mass density of plasma, ρ_{M}, and thus decrease the amplitude of ZFs; while for ZFs driven by main ion temperature perturbations, the presence of impurities can decrease the main ion mass density fraction, $\rho_{\mathrm{M,i}}/\rho_{\mathrm{M}}$, and thus decrease the amplitude of ZFs. To verify these results, a set of simulations is performed using the global gyrokinetic code NLT.[27,28]

A. Introduction to the NLT code

NLT is a δf gyrokinetic code, which solves the gyrokinetic Vlasov equation using the numerical Lie-transform method.[29–32] The early electrostatic version[27] of the NLT code with the adiabatic electron model has been verified by the linear GAM test and the standard Cyclone Base Case (CBC) for the linear ITG test, in which the magnetic flux coordinates (ψ, θ, ζ) are adopted, with ψ the poloidal magnetic flux, θ the poloidal angle, and ζ the toroidal angle. Then, the NLT code is upgraded[28] by adopting the Fourier filter and the field-aligned coordinates (ψ, α, θ) with $\alpha = q\theta - \zeta$. The nonlinear ITG turbulence test has been performed[28] for verifying the upgraded NLT code, and the results obtained by the NLT code agree well with those obtained by other gyrokinetic codes.

The gyrokinetic Vlasov equation used in the NLT code is given by

$$\frac{\mathrm{d}F_j}{\mathrm{d}t} = \frac{\partial F_j}{\partial t} + \dot{\boldsymbol{X}}\cdot\nabla F_j + \dot{v}_\parallel \frac{\partial F_j}{\partial v_\parallel} = 0. \tag{22}$$

Here, $F_j(\boldsymbol{X}, v_\parallel, \mu, t) = \delta F_j + F_{0,j}$ is the gyrocenter distribution function of j-species, with δF_j and $F_{0,j}$ being the perturbed and equilibrium parts of the distribution function, respectively. Since we focus on the ZF dynamics, the perturbed distribution function δF_j is axisymmetric in our simulation. $\dot{\boldsymbol{X}} = \{\boldsymbol{X}, H_{0,j} + H_{1,j}\}$ and $\dot{v}_\parallel = \{v_\parallel, H_{0,j} + H_{1,j}\}$, with $\{\cdot,\cdot\}$ denoting the Poisson bracket determined by the equilibrium fields. $H_{0,j} = m_j v_\parallel^2/2 + \mu B$ and $H_{1,j} = e_j\langle\delta\Phi\rangle_{\mathrm{GA}}$ are the gyrocenter equilibrium and perturbed Hamiltonian, respectively. $\delta\Phi$ is the perturbed electrostatic potential. In this work, the zonal part of $\delta\Phi$ is taken into consideration, which is denoted by $\delta\phi$. $\langle\cdot\rangle_{\mathrm{GA}}$ denotes the gyroaverage and is defined as

$$\langle A\rangle_{\mathrm{GA}}(\boldsymbol{X}, \mu) = \frac{1}{2\pi}\oint \mathrm{d}\xi A\left[\boldsymbol{X} + \boldsymbol{\rho}_{\mathrm{L},j}(\mu, \xi)\right], \tag{23}$$

with ξ being the gyroangle. In this work, the linear version of the NLT code is used to simulate the evolution of ZFs with the presence of impurities. Therefore, the nonlinear terms in Eq. (22) are ignored.

In the electrostatic adiabatic electron model, the zonal perturbed electrostatic potential $\delta\phi$ is solved by the quasi-neutrality equation in the long-wavelength limit, which is written as

$$\nabla\cdot\left(\sum_{j=\mathrm{i,z}}\frac{n_j m_j}{B^2}\nabla_\perp\delta\phi\right) - \frac{n_{\mathrm{e}} e_{\mathrm{e}}^2}{T_{\mathrm{e}}}\left(\delta\phi - \langle\delta\phi\rangle_{\mathrm{FA}}\right) = -\sum_{j=\mathrm{i,z}} e_j \delta n_{j,\mathrm{gy}}. \tag{24}$$

Here,

$$\delta n_{j,\mathrm{gy}} = 2\pi\int \mathrm{d}v_\parallel \mathrm{d}\mu \frac{B_\parallel^*}{m_j}\langle\delta F_j\rangle_{\mathrm{GA}} \tag{25}$$

denotes the ion perturbed gyrocenter density. $B_\parallel^* = \boldsymbol{b}\cdot\boldsymbol{B}^*$, with $\boldsymbol{B}^* = \boldsymbol{B} + \frac{m_j}{e_j} v_\parallel \nabla\times\boldsymbol{b}$.

B. Simulation setup

In order to simulate ZFs in the edge plasma, the pedestal parameters[33] in the DIII-D tokamak are chosen in this work, with the major/minor radius being $R_0/a = 1.67$ m/0.878 m. Note that, compared to the parameters presented in Ref. 33, the minor radius a in our simulations is expanded to ensure that ZFs away from the radial boundary of the simulation domain, while the absolute position of ZFs in our simulations is close to that of the pedestal in Ref. 33. The magnetic field at the magnetic axis is $B_0 = 2.1$ T. Impurity species is chosen as C^{6+}. The safety factor, equilibrium density, and temperature in our simulations are set as constants with $q = 3$, $n_e = 2\times10^{19}$ m^{-3}, $T_e = T_i = T_z = 0.18$ keV, respectively. Note that the equilibrium density of main ions n_i and impurity ions n_z satisfy the quasi-neutrality condition, $e_i n_i + e_z n_z = -e_e n_e$. The simulation domain is $r/a \in [0,1]$, $\alpha \in [0, 2\pi]$, $\theta \in [-\pi, \pi]$, $v_\parallel/c_{s,j} \in [-4, 4]$, $\mu B_0/T_j \in [0, 4^2/\sqrt{2}]$; here, $c_{s,j} = \sqrt{T_j/m_j}$ is the sound speed of the ion species j. The grid numbers are $N_r \times N_\alpha \times N_\theta \times N_{v_\parallel} \times N_\mu = 222 \times 4 \times 64 \times 64 \times 16$. μ is discretized according to the Gauss–Legendre formula, while the other variables are discretized uniformly. The push time step of simulation $\Delta t = 4\tau_{i,gy}$, with $\tau_{i,gy}$ being the main ion gyro-period at the magnetic axis. A damping buffer region near the radial outer boundary of the computational domain is adopted, and the perturbed potential on the radial outer boundary is set to zero. A mean value theorem[34] is used to obtain the perturbed distribution function and potential at the magnetic axis.

The initial ion density perturbation, $\delta n_j(0)$, and temperature perturbation, $\delta T_j(0)$, are set as follows:

$$[\delta n_j(0), \delta T_j(0)] = \begin{cases} A[n_j, T_j]\sin[k_r(r - r_1)] & \text{for } r \in [r_1, r_2], \\ 0 & \text{for } r \notin [r_1, r_2]. \end{cases} \tag{26}$$

Here, A is a small parameter which controls the amplitude of perturbations. $k_r = 2\pi/(r_2 - r_1)$ with $r_1 = 0.55a$ and $r_2 = 0.75a$, thus $k_r\rho_{L,i} \ll 1$. To investigate the effect of impurity concentration on ZFs, a set of effective charge numbers Z_{eff} are chosen and the electron equilibrium density n_e is fixed. Here, $Z_{\text{eff}} = \sum_{j=i,z} n_j Z_j^2/n_e$, with Z_j being the charge number of ion species j. Note that $Z_{\text{eff}} = 1$ in pure plasma, while Z_{eff} increases with the impurity concentration, and the increase in Z_{eff} leads to decrease in the main ion mass density fraction, $\rho_{M,i}/\rho_M$, in the admixed plasma. The amplitude of initial perturbations $\delta n_j(0)$ and $\delta T_j(0)$ are fixed in different Z_{eff}. In addition, to investigate the role of the main ion mass in the effect of impurities on ZFs, two main ion species, hydrogen H^+ and deuterium D^+, are chosen in our simulations. Collisional effects are not considered in this work, since we focus on the effects of impurities on ZFs in the absence of collisional damping. The CBC parameters[35] with non-flat equilibrium profiles are also used in our simulations, and the simulation results are presented in the Appendix.

C. Simulation results

1. Density perturbations

Figure 1 shows the effects of impurity ions C^{6+} on ZFs driven by main ion density perturbations in different Z_{eff}. As shown in Figs. 1(a1) and 1(a2), $\delta\phi(\infty)/\delta\phi(0)$ driven by the main ion density perturbations agrees with the theoretical prediction [Eq. (17)] and does not vary with Z_{eff}. This indicates that the presence of impurities has little effect on the neoclassical polarization shielding factor.

Figure 1(b1) shows that both the classical relative dielectric constant, c^2/V_A^2, and the initial zonal potential $\delta\phi(0)$ do not vary with Z_{eff} if the main ion is deuterium D^+. This is because the electron equilibrium density n_e is fixed and the charge-to-mass ratio of C^{6+} equals that of D^+, and thus the presence of C^{6+} does not affect the total ion mass density, ρ_M, as can be seen in Eq. (19). Since the amplitude of the main ion density perturbations, $\delta n_i(0)$, is fixed in different Z_{eff}, the value of the initial zonal potential $\delta\phi(0)$ is unchanged in different Z_{eff}, as can be seen in Eq. (16). These simulation results indicate that the amplitude of ZFs driven by the main ion density perturbations is not affected by impurities, if the electron equilibrium density n_e is fixed and the charge-to-mass ratio of the impurity ion equals that of the main ion.

However, Fig. 1(b2) shows that if the main ion is hydrogen H^+, whose charge-to-mass ratio is larger than that of C^{6+}, the presence of impurities increases the classical relative dielectric constant, c^2/V_A^2,

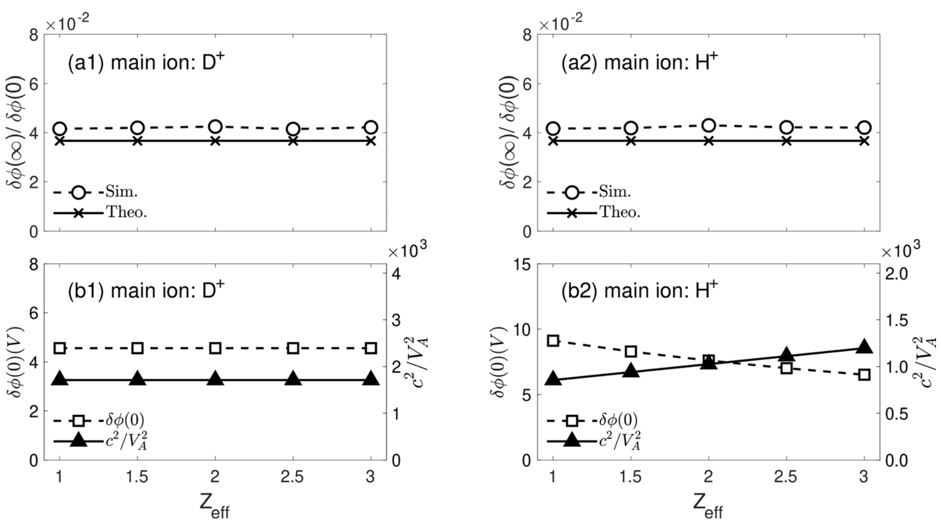

FIG. 1. ZFs driven by main ion density perturbations. The theoretical (Theo.) and simulation (Sim.) results of $\delta\phi(\infty)/\delta\phi(0)$ for different Z_{eff} when the main ion is deuterium D^+ (a1) or hydrogen H^+ (a2). Simulation results of the initial zonal potential $\delta\phi(0)$ and the value of the classical relative dielectric constant c^2/V_A^2 for different Z_{eff} when the main ion is deuterium D^+ (b1) or hydrogen H^+ (b2). Here, the radial position is $r = 0.6a$.

and decreases the value of the initial zonal potential $\delta\phi(0)$. Note that when $Z_{\rm eff}$ increases from 1 to 3, the classical relative dielectric constant, c^2/V_A^2, increases by nearly 40%, which is due to the fact that the total ion mass density, ρ_M, increases by 40%, as can be seen in Eq. (19); however, the value of the initial zonal potential $\delta\phi(0)$ decreases only by nearly 30%. This is due to the effects of adiabatic electrons, as can be seen in Eq. (24) used in this simulation, while the effects of adiabatic electrons are ignored in Eq. (16) to simplify the theoretical model. The simulation results in Fig. 1(b2) qualitatively verify the theoretical predictions [Eqs. (16) and (19)] that if the electron equilibrium density n_e is fixed and the charge-to-mass ratio of the impurity ion is smaller than that of the main ion, the presence of impurities can decrease the amplitude of ZFs, although it has little effect on the neoclassical polarization shielding factor.

2. Temperature perturbations

Figure 2 shows the effects of impurity ions C^{6+} on ZFs driven by ion temperature perturbations in the deuterium plasma. Note that the amplitudes of the temperature perturbations are fixed in all the following cases.

We first focus on ZFs driven by main ion temperature perturbations, $\delta T_i(0)$. From Fig. 2, it can be seen that the simulation results of the ZF amplitude for $Z_{\rm eff}=2$ (blue square symbols) are lower by nearly 20% than that for $Z_{\rm eff}=1$ (blue triangle symbols). This result is consistent with the theoretical prediction [Eq. (21)] that the ZF amplitude decreases by 20% from $Z_{\rm eff}=1$ (the blue solid line) to $Z_{\rm eff}=2$ (the blue dashed line). This simulation result verifies that the presence of impurities can decrease the amplitude of ZF driven by the main ion temperature perturbations.

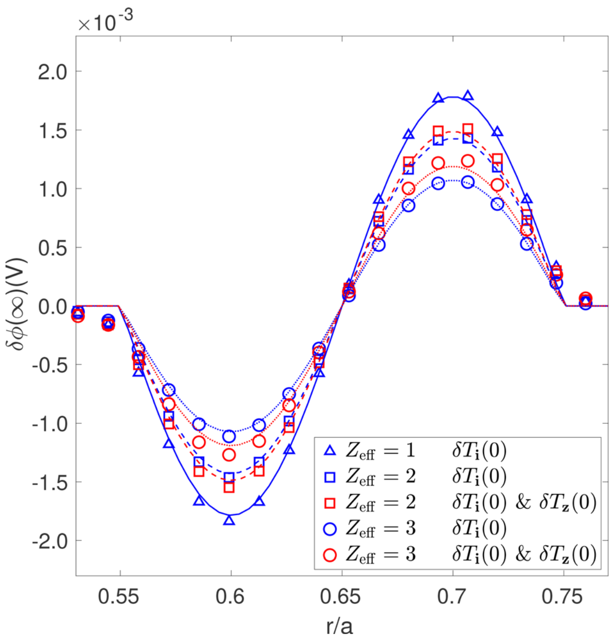

FIG. 2. ZFs driven by ion temperature perturbations for $Z_{\rm eff}=1$, 2, and 3, when the main ion is deuterium D^+. The blue symbols (lines) denote the simulation (theoretical) results of ZFs driven by the main ion temperature perturbations, $\delta T_i(0)$. The red symbols (lines) denote the simulation (theoretical) results of ZFs driven by the main ion together with the impurity ion temperature perturbations, $\delta T_i(0)$ and $\delta T_z(0)$. The radial domain [0.53a,0.77a] is presented here, because the initial perturbations are set to zero outside the range [0.55a,0.75a].

Then, when the impurity concentration increases (e.g., $Z_{\rm eff}$ increases from 2 to 3 in our simulation), it can be seen from Fig. 2 that the decreasing effect on the amplitude of ZF changes from a value of nearly 20% for $Z_{\rm eff}=2$ (by comparing the blue square symbols and the blue triangle symbols) to a value of nearly 40% for $Z_{\rm eff}=3$ (by comparing the blue circle symbols and the blue triangle symbols). This simulation result verifies that as the impurity concentration increases, the decreasing effects of impurities on the amplitude of ZF driven by the main ion temperature perturbations are enhanced.

Finally, we focus on ZFs driven by the main ion together with the impurity ion temperature perturbations, $\delta T_i(0)$ and $\delta T_z(0)$. Note that, the amplitude of $\delta T_z(0)$ equals that of $\delta T_i(0)$ in our simulation. From Fig. 2, it can be seen that when $Z_{\rm eff}=2$, the amplitude of ZF driven by $\delta T_i(0)$ and $\delta T_z(0)$ (red square symbols) is close to that only driven by $\delta T_i(0)$ (blue square symbols). The small deviation between these two amplitudes indicates the weak driving effects of impurity ion temperature perturbations on ZFs. This simulation result verifies that the driving effects of impurity ion temperature perturbations on ZFs can be ignored, when the amplitude of $\delta T_z(0)$ equals that of $\delta T_i(0)$. This conclusion does not change when $Z_{\rm eff}=3$, as can be seen from the red circle symbols and the blue circle symbols in Fig. 2.

IV. SUMMARY AND DISCUSSION

In summary, using gyrokinetic theory and simulations, we have investigated the effects of impurities on mesoscale ZFs in tokamak plasmas. It is found that ZFs can be driven by ion density and temperature perturbations, which represent the contributions from turbulent poloidal Reynolds stress[9] and turbulent energy flux,[10] respectively. For ZFs driven by main ion density perturbations, the presence of impurities can increase the total ion mass density of plasmas, and thus decrease the amplitude of ZFs; while it has little effect on the neoclassical polarization shielding factor. On the other hand, for ZFs driven by the main ion temperature perturbations, the presence of impurities can decrease the main ion mass density fraction and, thus, decrease the amplitude of ZFs. In deuterium plasma with impurity ion C^{6+}, it is found that the amplitude of ZF driven by main ion temperature perturbations decreases by nearly 40% for $Z_{\rm eff}$ varying from 1 to 3. Finally, the driving effects of impurity ion temperature perturbations on ZFs can be ignored, due to the low concentration of impurity ions.

Previous gyrokinetic theory,[16] which considers ZFs driven by initial charge density perturbations, shows that the presence of impurities has very weak influence on the residual ZF in the long-wavelength limit. In this work, we point out that the presence of impurities has significant decreasing effects on the amplitude of ZFs driven by main ion temperature perturbations. Furthermore, this finding might provide an explanation for the phenomenon observed in Ref. 18 that the time-averaged absolute value of zonal radial electric fields is smaller when impurity is present.

The results presented in this paper regarding the effects of impurities on ZFs are obtained based on the Rosenbluth–Hinton collisionless gyrokinetic model,[5] which is similar to the approach used in previous works;[16–18] collisional effects are ignored here, similar to Refs. 5 and 16–18. However, it should be pointed out that ion–ion collision can reduce the amplitude of ZFs in a pure plasma[22] or in a plasma with impurity.[15] It is found that the ratio of the amplitude of collisional damped zonal potential to the initial zonal potential $\phi_k(\infty)/\phi_k(0)$ approaches B_p^2/B^2 in the long-time limit in a pure plasma,[22] with B_p being the poloidal component of the magnetic field. This ratio is regardless of whether impurities are present or not.[15] Nevertheless, it is found that impurities do accelerate collisional damping since they increase the effective collisionality of the main ions.[15] In future D–T fusion reactors, plasma is collisionless. Therefore, the main result of this paper, Eq. (15), should be understood as the upper limit for the amplitude of ZFs driven by ITG turbulence in plasmas with impurities, providing a benchmark for turbulence simulations of such plasmas.

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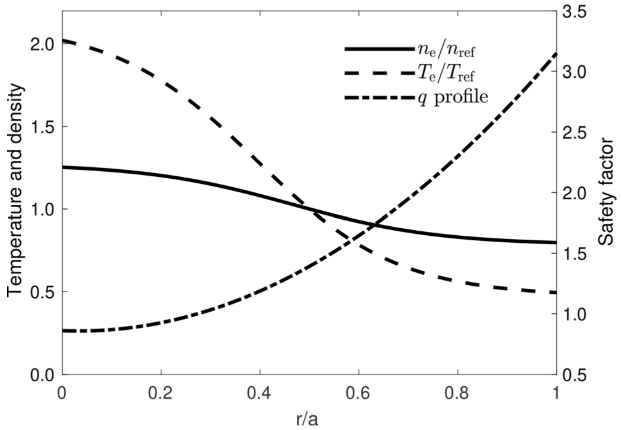

FIG. 3. Equilibrium profiles under the CBC parameters. Electron density and temperature profiles are referenced to the left-hand axis, while the safety factor profile is referenced to the right-hand axis.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Yuesong Li and Zihao Wang contributed equally to this paper.

Yuesong Li: Data curation (lead); Investigation (equal); Methodology (equal); Software (equal); Writing – original draft (equal). **Zihao Wang:** Methodology (equal); Software (equal); Writing – original draft (equal); Writing – review & editing (equal). **Tiannan Wu:** Investigation (equal); Methodology (equal); Software (equal). **Yuefeng Qiu:** Formal analysis (supporting); Software (supporting); Writing – review & editing (supporting). **Jie Wang:** Formal analysis (supporting); Writing – review & editing (supporting). **Shaojie Wang:** Conceptualization (equal); Methodology (equal); Supervision (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

APPENDIX: SIMULATION RESULTS WITH THE CBC PARAMETERS

To verify the conclusions for non-flat equilibrium profiles, ZFs in the presence of impurities are simulated with the CBC parameters. The simulation results show that the qualitative conclusions remain unchanged for non-flat equilibrium profiles.

The CBC parameters are as follows: the major/minor radius is $R_0/a = 1.67$ m/0.60 m and the magnetic field at the magnetic axis is $B_0 = 1.9$ T. The electron equilibrium density, temperature, and safety factor profiles are shown in Fig. 3, with $n_{\rm ref} = 10^{19}$ m^{-3} and $T_{\rm ref} = 1.97$ keV. The impurity species is still chosen as C^{6+}. The equilibrium density of main ions $n_{\rm i}$ and impurity ions $n_{\rm z}$ satisfies the quasi-neutrality condition, $e_{\rm i}n_{\rm i} + e_{\rm z}n_{\rm z} = -e_{\rm e}n_{\rm e}$. The temperature profiles are the same for all ion species and electrons, i.e., $T_{\rm i} = T_{\rm z} = T_{\rm e}$. The simulation

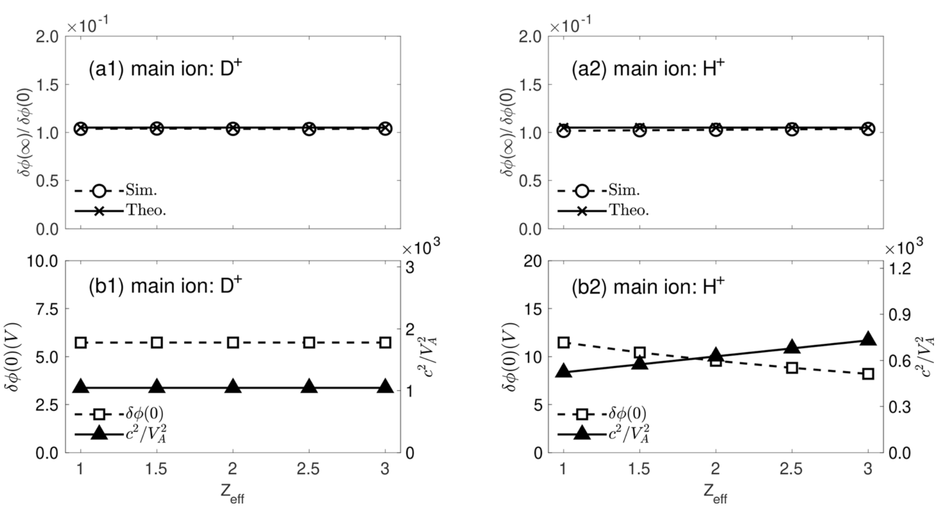

FIG. 4. ZFs driven by the main ion density perturbations with the CBC parameters. Theoretical and simulation results of $\delta\phi(\infty)/\delta\phi(0)$ for different $Z_{\rm eff}$ when the main ion is deuterium D^+ (a1) or hydrogen H^+ (a2). Simulation results of the initial zonal potential $\delta\phi(0)$ and the value of the classical relative dielectric constant c^2/V_A^2 for different $Z_{\rm eff}$ when the main ion is deuterium D^+ (b1) or hydrogen H^+ (b2). Here, the radial position $r = 0.525a$.

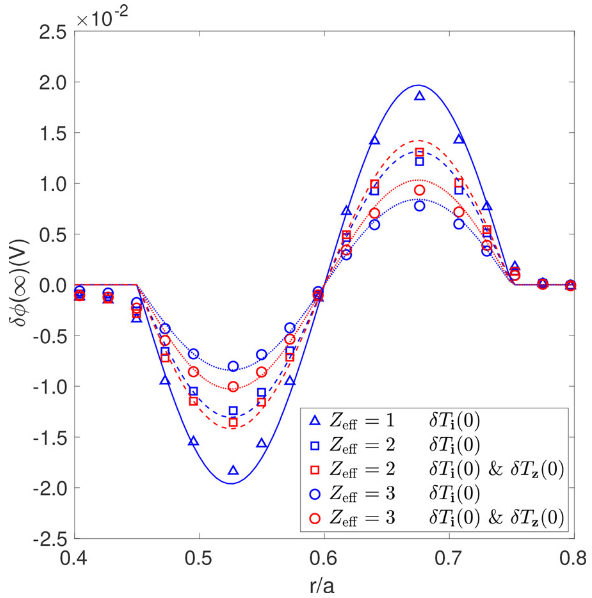

FIG. 5. ZFs driven by ion temperature perturbations for $Z_{\rm eff}=1$, 2, and 3 with the CBC parameters. The blue symbols (lines) denote the simulation (theoretical) results of ZFs driven by main ion temperature perturbations $\delta T_{\rm i}(0)$. The red symbols (lines) denote the simulation (theoretical) results of ZFs driven by the main ion together with the impurity ion temperature perturbations $\delta T_{\rm i}(0)$ and $\delta T_{\rm z}(0)$. The radial domain [0.4a,0.8a] is presented here, because the initial perturbations are set to zero outside the range [0.45a,0.75a].

domain and grid numbers are the same as that presented in Sec. III B. The initial ion density and temperature perturbations are still set as Eq. (26), except for $r_1=0.45a$ and $r_2=0.75a$, to ensure $k_r\rho_{\rm L,i}\ll 1$.

The simulation results of ZFs driven by ion density and temperature perturbations with the CBC parameters are shown in Figs. 4 and 5, respectively. It can be seen that the qualitative conclusions remain unchanged for non-flat equilibrium profiles. Note that there are some discrepancies between the simulation results and the theoretical predictions in Fig. 5. The discrepancies are attributed to the finite orbit width effect, which is represented by $Q_{\rm i}=k_r(r-\overline{r})$ in Eq. (13). $Q_{\rm i}\ll 1$ is used to analytically obtain the asymptotic zonal potential, while $Q_{\rm i}\approx 0.35$ in the simulation with CBC parameters. The finite orbit width effect will flatten the radial distribution of the temperature perturbations and, thus, cause discrepancies. As $Q_{\rm i}\approx 0.28$ in the simulation with pedestal parameters in Sec. III C, the discrepancies between the simulation results and the theoretical predictions in Fig. 2 are smaller compared to those in Fig. 5.

REFERENCES

[1] H. Biglari, P. H. Diamond, and P. W. Terry, Phys. Fluids B **2**(1), 1–4 (1990).
[2] K. H. Burrell, Phys. Plasmas **4**, 1499 (1997).
[3] Z. Lin, T. S. Hahm, W. W. Lee, W. M. Tang, and R. B. White, Science **281**, 1835 (1998).
[4] F. Wagner, G. Becker, K. Behringer, D. Campbell, A. Eberhagen, W. Engelhardt, G. Fussmann, O. Gehre, J. Gernhardt, G. V. Gierke, G. Haas, M. Huang, F. Karger, M. Keilhacker, O. Klüber, M. Kornherr, K. Lackner, G. Lisitano, G. G. Lister, H. M. Mayer, D. Meisel, E. R. Müller, H. Murmann, H. Niedermeyer, W. Poschenrieder, H. Rapp, H. Röhr, F. Schneider, G. Siller, E. Speth, A. Stäbler, K. H. Steuer, G. Venus, O. Vollmer, and Z. Yü, Phys. Rev. Lett. **49**, 1408 (1982).
[5] M. N. Rosenbluth and F. L. Hinton, Phys. Rev. Lett. **80**, 724 (1998).
[6] L. Chen, Z. Lin, and R. White, Phys. Plasmas **7**, 3129 (2000).
[7] P. H. Diamond, Y.-M. Liang, B. A. Carreras, and P. W. Terry, Phys. Rev. Lett. **72**, 2565 (1994).
[8] E.-J. Kim and P. H. Diamond, Phys. Rev. Lett. **90**, 185006 (2003).
[9] D. Zhang, Y. Xu, Z. Dai, and S. Wang, Nucl. Fusion **60**, 046015 (2020).
[10] S. Wang, Phys. Plasmas **24**, 102508 (2017).
[11] C. Ajay, S. Brunner, B. McMillan, J. Ball, J. Dominski, and G. Merlo, J. Plasma Phys. **86**, 905860504 (2020).
[12] Z. Wang, Z. Dai, and S. Wang, Phys. Rev. E **106**, 035205 (2022).
[13] Z. Wang and S. Wang, Nucl. Fusion **65**, 016034 (2025).
[14] J. Wesson, *Tokamaks*, 4th ed. (Clarendon Press, Oxford, 2011), Chap. 9.
[15] S. Braun, P. Helander, E. A. Belli, and J. Candy, Plasma Phys. Controlled Fusion **51**, 065011 (2009).
[16] W. Guo, L. Wang, and G. Zhuang, Nucl. Fusion **57**, 056012 (2017).
[17] X. Wei, H. Yang, S. Li, and Y. Xiao, Phys. Plasmas **25**, 082508 (2018).
[18] J. Seo, H. Jhang, and J.-M. Kwon, Phys. Plasmas **29**, 052502 (2022).
[19] A. R. Knyazev and S. I. Krasheninnikov, Phys. Plasmas **31**, 012502 (2024).
[20] J. Cheng, J. Q. Dong, K. Itoh, L. W. Yan, M. Xu, K. J. Zhao, W. Y. Hong, Z. H. Huang, X. Q. Ji, W. L. Zhong, D. L. Yu, S.-I. Itoh, L. Nie, D. F. Kong, T. Lan, A. D. Liu, X. L. Zou, Q. W. Yang, X. T. Ding, X. R. Duan, Y. Liu *et al.*, Phys. Rev. Lett. **110**, 265002 (2013).
[21] J. Cheng, J. Q. Dong, K. Itoh, L. W. Yan, W. Y. Hong, K. J. Zhao, Z. H. Huang, X. Q. Ji, W. L. Zhong, D. L. Yu, S.-I. Itoh, L. Nie, X. M. Song, Q. W. Yang, X. T. Ding, X. L. Zou, X. R. Duan, Y. Liu *et al.*, Nucl. Fusion **54**, 114004 (2014).
[22] F. L. Hinton and M. N. Rosenbluth, Plasma Phys. Controlled Fusion **41**, A653 (1999).
[23] E. A. Frieman and L. Chen, Phys. Fluids **25**, 502 (1982).
[24] W. Guo, S. Wang, and J. Li, Phys. Plasmas **17**, 112510 (2010).
[25] L. Ye, W. Guo, X. Xiao, and S. Wang, Phys. Plasmas **20**, 072501 (2013).
[26] B. A. Grierson, K. H. Burrell, C. Chrystal, R. J. Groebner, D. H. Kaplan, W. W. Heidbrink, J. M. Muñoz Burgos, N. A. Pablant, W. M. Solomon, and M. A. Van Zeeland, Rev. Sci. Instrum. **83**, 10D529 (2012).
[27] L. Ye, Y. Xu, X. Xiao, Z. Dai, and S. Wang, J. Comput. Phys. **316**, 180 (2016).
[28] Y. Xu, L. Ye, Z. Dai, X. Xiao, and S. Wang, Phys. Plasmas **24**, 082515 (2017).
[29] S. Wang, Phys. Plasmas **19**, 062504 (2012).
[30] S. Wang, Phys. Rev. E **87**, 063103 (2013).
[31] S. Wang, Phys. Plasmas **20**, 082312 (2013).
[32] S. Wang, Phys. Plasmas **21**, 072312 (2014).
[33] K. H. Burrell, T. N. Carlstrom, E. J. Doyle, D. Finkenthal, P. Gohil, R. J. Groebner, D. L. Hillis, J. Kim, H. Matsumoto, R. A. Moyer, T. H. Osborne, C. L. Rettig, W. A. Peebles, T. L. Rhodes, H. StJohn, R. D. Stambaugh, M. R. Wade, and J. G. Watkins, Plasma Phys. Controlled Fusion **34**, 1859 (1992).
[34] T. Wu, Z. Wang, and S. Wang, arXiv: 2402.03657 (2025).
[35] A. M. Dimits, G. Bateman, M. A. Beer, B. I. Cohen, W. Dorland, G. W. Hammett, C. Kim, J. E. Kinsey, M. Kotschenreuther, A. H. Kritz, L. L. Lao, J. Mandrekas, W. M. Nevins, S. E. Parker, A. J. Redd, D. E. Shumaker, R. Sydora, and J. Weiland, Phys. Plasmas **7**, 969 (2000).